

PUMP TROUBLESHOOTING AND DIAGNOSTIC GUIDE

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Asset Class: Centrifugal Pump

Purpose: Systematic Diagnosis, Root Cause Identification and Corrective Action Guidance

1. Purpose

This troubleshooting guide provides a structured methodology for diagnosing abnormal pump behavior using:

- Telemetry Data
- AI Predictions
- Operational Observations
- Reliability Principles

This document is intended for:

- Control Room Operators
 - Maintenance Engineers
 - Reliability Engineers
 - Industrial AI Copilot
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2. Diagnostic Philosophy

Effective troubleshooting requires:

Step 1: Identify abnormal telemetry.

Step 2: Identify affected subsystem.

Step 3: Determine probable failure mode.

Step 4: Verify diagnosis using supporting telemetry.

Step 5: Determine urgency.

Step 6: Recommend corrective action.

3. High Vibration Troubleshooting

Symptom

Telemetry:

vibration > 4 mm/s

Step 1

Check Bearing Temperature

Telemetry:

bearing_temp

Decision:

If bearing temperature is increasing:

Proceed to Bearing Failure Investigation.

Otherwise:

Continue.

Step 2

Check Suction Pressure

Telemetry:

suction_pressure

Decision:

If suction pressure is decreasing:

Proceed to Cavitation Investigation.

Otherwise:

Continue.

Step 3

Check Motor Current

Telemetry:

motor_current

Decision:

If current is increasing:

Mechanical resistance likely.

Possible Causes:

- Bearing degradation
 - Misalignment
 - Rotor imbalance
-

Step 4

Review Historical Trend

Evaluate:

- Previous 7 days
- Previous 30 days

Trend Categories:

Stable

Gradual Increase

Rapid Increase

Intermittent Increase

AI Diagnostic Logic

If:

Vibration ↑

Bearing Temperature ↑

Motor Current ↑

Diagnosis:

Bearing Failure

Confidence:

90%+

If:

Vibration ↑

Suction Pressure ↓

Flow Rate ↓

Diagnosis:

Cavitation

Confidence:

90%+

4. High Bearing Temperature Troubleshooting

Symptom

Telemetry:

```
bearing_temp > 75°C
```

Step 1

Check Lubrication Level

Telemetry:

```
lubrication_level
```

If:

Lubrication ↓

Probable Cause:

Lubrication deficiency

Step 2

Check Vibration

Telemetry:

```
vibration
```

If:

Vibration ↑

Probable Cause:

Bearing wear

Step 3

Check Motor Current

Telemetry:

motor_current

If:

Current ↑

Probable Cause:

Mechanical drag

Diagnosis Matrix

Temperature	Vibration	Current	Likely Cause
High	Normal	Normal	Lubrication Issue
High	High	Normal	Bearing Wear
High	High	High	Advanced Bearing Failure
High	Normal	High	Mechanical Resistance

5. Reduced Flow Rate Troubleshooting

Symptom

Telemetry:

flow_rate < 250 m³/hr

Step 1

Check Suction Pressure

If:

Suction Pressure ↓

Possible Cause:

Cavitation

Step 2

Check Seal Leakage

Telemetry:

seal_leakage_rate

If:

Leakage ↑

Possible Cause:

Seal Failure

Step 3

Check Discharge Pressure

If:

Discharge Pressure ↓

Possible Causes:

- Seal Leakage
- Internal Wear

- Impeller Damage
-

AI Diagnostic Logic

Flow ↓

Discharge Pressure ↓

Seal Leakage ↑

Diagnosis:

Seal Failure

Confidence:

High

Flow ↓

Suction Pressure ↓

Vibration ↑

Diagnosis:

Cavitation

Confidence:

High

6. Seal Leakage Troubleshooting

Symptom

Telemetry:

```
seal_leakage_rate > 0.1 L/hr
```

Step 1

Determine Leakage Trend

Categories:

Stable

Increasing

Rapid Increase

Step 2

Check Vibration

If:

Vibration ↑

Possible Cause:

Misalignment

Bearing degradation

Step 3

Check Discharge Pressure

If:

Pressure ↓

Likely Cause:

Seal deterioration

Step 4

Inspect Seal Assembly

Inspect:

- Seal Faces
 - O-rings
 - Shaft Sleeve
-

AI Diagnostic Logic

Leakage ↑

Pressure ↓

Flow ↓

Diagnosis:

Seal Failure

Confidence:

95%

7. Cavitation Investigation

Symptom

Operator Reports:

- Crackling noise
- Gravel sound
- Flow instability

Telemetry:

- Low Suction Pressure
 - Increased Vibration
 - Reduced Flow
-

Step 1

Verify Suction Pressure

Telemetry:

suction_pressure

If:

< 2 bar

High cavitation risk

Step 2

Inspect Strainers

Check:

- Blockage
 - Fouling
-

Step 3

Inspect Valves

Verify:

- Valve position
 - Partial closure
-

Step 4

Verify NPSH

Calculate:

Available NPSH

Required NPSH

AI Diagnostic Logic

Low Suction Pressure

+

High Vibration

+

Flow Reduction

=

Cavitation

Confidence:

95%

8. AI Copilot Escalation Logic

Low Risk

Failure Probability:

< 50%

Action:

Monitor

Medium Risk

Failure Probability:

50% – 80%

Action:

Maintenance Notification

High Risk

Failure Probability:

80% – 90%

Action:

Schedule Maintenance

Critical Risk

Failure Probability:

90% – 95%

RUL < 30 Days

Action:

Reliability Engineer Review

Emergency

Failure Probability:

95%

RUL < 10 Days

Action:

Immediate Maintenance Planning

9. Example Diagnostic Cases

Case 1

Telemetry:

Bearing Temp:

89°C

Vibration:

8.4 mm/s

Motor Current:

142 A

Failure Probability:

93%

RUL:

14 Days

Diagnosis:

Advanced Bearing Failure

Recommended Action:

Bearing replacement within 48 hours

Case 2

Telemetry:

Seal Leakage:

0.9 L/hr

Flow Rate:

190 m³/hr

Discharge Pressure:

7 bar

Failure Probability:

91%

RUL:

18 Days

Diagnosis:

Severe Seal Failure

Recommended Action:

Seal replacement

Case 3

Telemetry:

Suction Pressure:

1.6 bar

Flow Rate:

180 m³/hr

Vibration:

8.1 mm/s

Failure Probability:

89%

RUL:

22 Days

Diagnosis:

Severe Cavitation

Recommended Action:

Immediate suction system investigation

10. Summary

The most accurate troubleshooting approach combines:

- Telemetry Trends
- Failure Probability
- Failure Mode Classification
- Remaining Useful Life
- Engineering Inspection

The Industrial AI Copilot should always verify diagnoses using multiple telemetry indicators before recommending maintenance actions.